

Last Time:

- Def: V v.s. Then $W \subseteq V$ is a subsp of V if W w/ induced $+$ & $\cdot$ from V is a v.s.
- Subsp Crit: V v.s., $W \subseteq V$. Then W is a subsp of V iff
 - (1) $0_V \in W$
 - (2) $x + y \in W \quad \forall x, y \in W$
 - (3) $c \cdot x \in W \quad \forall c \in \mathbb{R} \text{ and } \forall x \in W$

Today:

- (1) Continue §1.2 Vector Spaces and Subspaces
- (2) Start §1.3 Linear Combinations, Span, and Linear Independence

FIS calls it $P(x)$ (very outdated)

Ex 1.24: Let $\mathbb{R}[x] \subseteq \mathcal{F}(\mathbb{R}, \mathbb{R})$ be the set of $(\mathbb{R})$ -^{polys}polynomials. I.e., $(\text{coeffs in } \mathbb{R})$

variable

$$\mathbb{R}[x] = \left\{ \sum_{i=0}^d a_i x^i : d \in \mathbb{Z}_{\geq 0}, a_i \in \mathbb{R} \quad \forall 0 \leq i \leq d \right\}$$

$\mathbb{Z}_{\geq 0} = \{0, 1, 2, \dots\}$ set of nonnegative integers

$a_d x^d + \dots + a_0 x^0$

is a fcn where $\forall c \in \mathbb{R}, c \mapsto a_d c^d + \dots + a_1 c^1 + a_0$

Then $\mathbb{R}[x]$ is a subsp of $\mathcal{F}(\mathbb{R}, \mathbb{R})$ by Subsp Crit.

Pf: (1): $0_{\mathcal{F}(\mathbb{R}, \mathbb{R})}$ is equal to $0 = 0 \cdot x^0 \in \mathbb{R}[x]$ b/c

$$0 = 0 \cdot x^0 : \mathbb{R} \rightarrow \mathbb{R} \quad \text{and} \quad 0_{\mathcal{F}(\mathbb{R}, \mathbb{R})} : \mathbb{R} \rightarrow \mathbb{R} \\ c \mapsto 0 \cdot c^0 = 0 \quad c \mapsto 0 \quad \square$$

(2): $\forall \sum_{\bar{i}=0}^{d_1} a_i x^{\bar{i}}, \sum_{\bar{i}=0}^{d_2} b_i x^{\bar{i}} \in \mathbb{R}[x]$, let $d := \max(d_1, d_2)$,

$a_j := 0 \quad \forall d_1 < j \leq d$, and $b_j := 0 \quad \forall d_2 < j \leq d$. Then

$$\sum_{\bar{i}=0}^{d_1} a_i x^{\bar{i}} + \sum_{\bar{i}=0}^{d_2} b_i x^{\bar{i}} = \sum_{\bar{i}=0}^d a_i x^{\bar{i}} + \sum_{\bar{i}=0}^d b_i x^{\bar{i}} = \sum_{\bar{i}=0}^d \underbrace{(a_i + b_i)}_{\mathbb{R}} x^{\bar{i}} \\ \in \mathbb{R}[x]. \quad \square$$

(3): $\forall c \in \mathbb{R} \quad \& \quad \forall \sum_{\bar{i}=0}^d a_i x^{\bar{i}} \in \mathbb{R}[x]$, then

$$c \cdot \sum_{\bar{i}=0}^d a_i x^{\bar{i}} = \sum_{\bar{i}=0}^d \underbrace{(c a_i)}_{\mathbb{R}} x^{\bar{i}} \in \mathbb{R}[x]. \quad \square$$

Ex 1.25: Given $f = \sum_{\bar{i}=0}^d a_i x^{\bar{i}} \in \mathbb{R}[x] \setminus \{0\}$ (so $f \neq 0$),

the degree of f is def as $\left(\begin{array}{l} \text{convention:} \\ a_j = 0 \quad \forall j > d \end{array} \right)$

$\deg(f) := \max \{ i \in \mathbb{Z}_{\geq 0} : a_i \neq 0 \}$. Let $\deg(0) := -\infty$,

treated as < 0 . $\forall n \in \mathbb{Z}_{\geq 0}$ fixed, def

$$\mathbb{R}[x]_{\leq n} := \left\{ \sum_{\hat{i}=0}^n a_{\hat{i}} x^{\hat{i}} : a_{\hat{i}} \in \mathbb{R} \quad \forall 0 \leq \hat{i} \leq n \right\} \subsetneq \mathbb{R}[x]$$

as the subset of $\deg \leq n$ polys ($0 \in \mathbb{R}[x]_{\leq n}$).

Then $\mathbb{R}[x]_{\leq n} \subsetneq \mathbb{R}[x]$ is a subsp.

Pf: Exercise by using Subsp Crit.

$X \subsetneq Y$
means $X \subseteq Y$
but $X \neq Y$

$$\left(\begin{array}{l} \text{Hint: use } \sum_{\hat{i}=0}^n, \text{ or use} \\ \deg(f+g) \leq \max(\deg f, \deg g) \ \& \\ \deg(fg) = \deg(f) + \deg(g) \text{ if } f, g \neq 0 \end{array} \right)$$

□

Ex 1.26 (MATH 161):

$C(\mathbb{R}, \mathbb{R}) := \{f \in \mathcal{F}(\mathbb{R}, \mathbb{R}) : f \text{ continuous on } \mathbb{R}\} \overset{\text{Calc I}}{\supset} \mathbb{R}[x]$
 $\subseteq \mathcal{F}(\mathbb{R}, \mathbb{R})$ subsp by Subsp Crit.

Exercise 1.27: Prove that each of the following subset is a subsp:

(a) $f \in \mathcal{F}(\mathbb{R}, \mathbb{R})$ is called an even fcn if
 $f(x) = f(-x) \quad \forall x \in \mathbb{R}$. Consider
 $\{f \in \mathcal{F}(\mathbb{R}, \mathbb{R}) : f \text{ even}\} \subseteq \mathcal{F}(\mathbb{R}, \mathbb{R})$.

(b) Fix any $a \in \mathbb{R}$ and $n \in \mathbb{Z}_{\geq 0}$. Consider

$$\{f \in \mathbb{R}[x]_{\leq n} : f(a) = 0\} \subseteq \mathbb{R}[x]_{\leq n}.$$

(c) Fix $n \in \mathbb{Z}_{>0}$. Square matrix $A \in M_n(\mathbb{R})$ is called symmetric if $A_{ij} = A_{ji} \quad \forall 1 \leq i, j \leq n$.

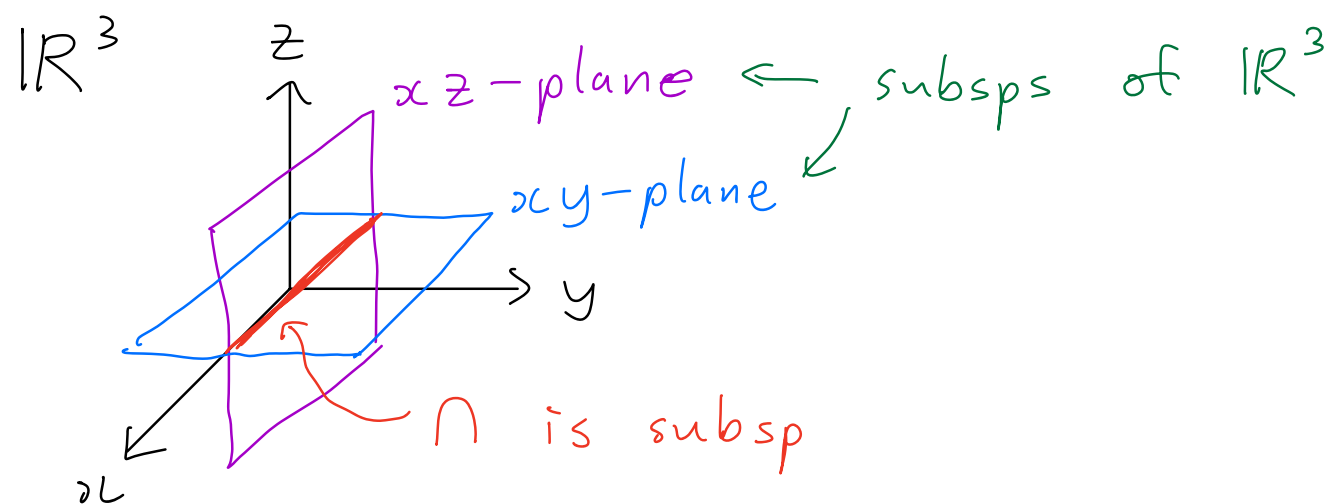
for example

$$\left(\begin{array}{l} \text{E.g.} \\ \text{sym} \end{array} \begin{bmatrix} 1 & 2 & -5 \\ 2 & 7 & 3 \\ -5 & 3 & 0 \end{bmatrix} \text{ sym}, \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix} \text{ not sym} \right)$$

Consider $\{A \in M_n(\mathbb{R}) : A \text{ sym}\} \subseteq M_n(\mathbb{R})$.

Recall: Ex 1.23 says $\exists$ subsp W_1, W_2 of a v.s. V s.t. $W_1 \cup W_2 \subseteq V$ not a subsp.

Q: $W_1 \cap W_2 \subseteq V$ subsp?



Prop 1.28: V v.s., $W_1, W_2 \subseteq V$ subsp. Then $W_1 \cap W_2$ is also a subsp of V .

Rmk 1.29: V v.s. Using Prop 1.28, we get:

(1) If $W_1, \dots, W_m \subseteq V$ subspaces, then so is

$W_1 \cap \dots \cap W_m$. (Pf: Exercise via Induction or (2))

(2) The power set of V is $\mathcal{P}(V) := \{Y : Y \subseteq V\}$, which is the set of all subsets of V . If

$\mathcal{C} \subseteq \mathcal{P}(V)$, then

$$\bigcap_{W \in \mathcal{C}} W := \{x \in V : x \in W \ \forall W \in \mathcal{C}\}$$

is a subspace of V . (Pf: FIS Thm 1.4)

Pf of Prop 1.28:

It suffices to check Subsp Crit on $W_1 \cap W_2$.

(1): $W_1, W_2 \subseteq V$ subspaces

$\Rightarrow 0_V \in W_1$ & $0_V \in W_2$ by Subsp Crit

$\Rightarrow 0_V \in W_1 \cap W_2$. ▢

(2): $\forall x, y \in W_1 \cap W_2$, note that $W_1 \cap W_2 \subseteq W_i \ \forall i$

$\Rightarrow x, y \in W_i \ \forall i$. Subsp Crit on W_1, W_2

$\Rightarrow x+y \in W_1, W_2 \Rightarrow x+y \in W_1 \cap W_2$. ▢

(3): $\forall c \in \mathbb{R}$ and $\forall x \in W_1 \cap W_2$, $W_1 \cap W_2 \subseteq W_i \ \forall i$

$\Rightarrow x \in W_i \ \forall i$. So $cx \in W_1, W_2$ by Subsp Crit

$$\Rightarrow cx \in W_1 \cap W_2.$$

□

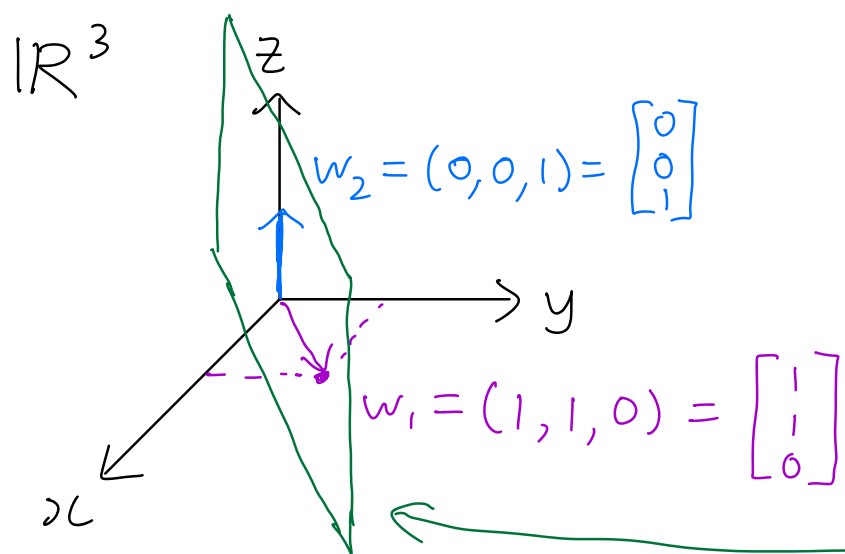
§1.3 Linear Combinations, Span, and Linear Independence

Motivation: We want to construct more subspaces.

Given v.s. V and $w_1, \dots, w_m \in V$, how to make a subspace W that contains $w_1, \dots, w_m$?

(Imagine this as a skeleton of W)

⇓



In this case, "visually,"

guess that $W = \mathbb{R}^3$ or $\left(\begin{array}{l} \text{Exercise:} \\ \text{Check } W \text{ is} \\ \text{a subspace} \end{array} \right)$

$$W = \left\{ \begin{bmatrix} a \\ a \\ b \end{bmatrix} : a, b \in \mathbb{R} \right\}$$

$$a \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} =$$

⇓

Def 1.30: V v.s.

(1) $v \in V$ is a linear combination of $w_1, \dots, w_k \in V$

if $\exists a_1, \dots, a_k \in \mathbb{R}$ s.t. $v = \sum_{i=1}^k a_i w_i$.

coeff of l.c.

(2) $v \in V$ is a l.c. of S (where $\emptyset \neq S \subseteq V$) if
 $\exists$ ^{fin} finite subset $\emptyset \neq T = \{w_1, \dots, w_k\} \subseteq S$ s.t. v is
 a l.c. of $w_1, \dots, w_k$.

(3) $v \in V$ is a l.c. of $\emptyset \subseteq V$ if $v = 0$.

Def 1.3: V v.s

(1) The span of $w_1, \dots, w_k \in V$ is a subset
 $\text{span}(w_1, \dots, w_k) := \{v \in V : v \text{ is a l.c. of } w_1, \dots, w_k\}$
 $= \left\{ \sum_{i=1}^k a_i w_i : a_i \in \mathbb{R} \ \forall 1 \leq i \leq k \right\} \subseteq V$

(2) The span of subset S of V is a subset
 $\text{span}(S) := \{v \in V : v \text{ is a l.c. of } S\} \subseteq V$.